

MSE 311 Thermodynamics of materials-Homework 4

Name:

Student ID:

1. To ease the energy crisis, the idea of an air-powered car has been mentioned repeatedly. Assuming the current efficiency of gasoline engine is about 36%, if a steel cylinder that can withstand pressure of 100 atm is used in the air-powered car, what size of the steel cylinder is needed to make the air-powered car run at the same distance as gasoline engine car with full tank (24 liters). Assuming the efficiency of air-powered car is 100%. Ignore the mass of the steel cylinder. (The calorific value of gasoline is 33580 kJ/liter. The ambient temperature is 300K.)

2. One mole of a monatomic ideal gas undergoes a reversible expansion at constant pressure during which the entropy of the gas increases by 14.41 J/K and the gas absorbs 6236 joules of heat. Calculate the initial and final temperatures of the gas. One mole of a second monatomic ideal gas undergoes a reversible isothermal expansion during which it doubles its volume, performs 1729 joules of work and increases its entropy by 5.763 J/K. Calculate the temperature at which the expansion was conducted.

3. Open Question. 1. Correcting the Otto cycle, , if there is a heat leak.